Socratic Steering: Adaptive Questioning Reveals Directional Influence and Resistant Opinion Phenotypes

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Abstract—Can strategically selected questions steer measured political opinion trajectories without presenting persuasive messages or prescribing answers? We introduce *Digital Twin Anchored Generation* (DTAG), a framework for adaptive inquiry that uses conditional dependencies inferred from survey responses to select questions by their expected directional effect. We construct a conditional generative model of the 2024 General Social Survey by estimating response dependencies across the full set of survey items represented in the model, without preselecting ideological topics, imposing predefined relationships, or reducing opinions to a few latent dimensions. The inferred conditional distributions induce a distance between response states and allow DTAG to select questions by their expected signed displacement toward *a priori* fixed opposing reference profiles. Importantly, DTAG selects only the questions; human participants supply every answer. Among 71 completed, baseline-calibrated human sessions, 59 received 20 directionally selected questions. Left push yielded leftward displacement in 20/21 participants (95.2%), while Right push yielded rightward displacement in 21/38 (55.3%). Movement could oppose baseline orientation: 4/4 initially Right-leaning participants moved left under Left push, and 12/29 initially Left-leaning participants moved right under Right push. The observed response phenotypes were PH1 (directionally responsive, 41/59), PH2 (Right-push resistant from a Left baseline, 17/29), and PH3 (Left-push resistant from a Left baseline, 1/17). The four-question rule flagged all 17 eventual PH2 outcomes, with two false positives. Supporting computational tests of 960 model-generated trajectories and 2,880 fixed-set replays confirmed directional selection across tested question orders. These findings reveal distinct, readily identifiable opinion-response phenotypes: individuals whose measured profiles move in the selected direction and individuals resistant to a given directional push. In this cohort, Right-push resistance was detectable from only four answers, demonstrating the potential for rapid identification of direction-specific resistance through adaptive questioning alone.

Index Terms—Predictive phenotypes, large science models, conditional response modeling, digital twins, survey methodology, opinion dynamics, adaptive questioning, generative simulation.

1 Introduction

Can asking a question influence the answers a person subsequently gives? Survey instruments are commonly treated as ways to measure opinions, but inquiry itself may affect which considerations become salient and how later responses are expressed. Effects of question wording and order have long been recognized [1], [2]; more generally, *reflexivity* raises the possibility that a measurement process contributes to the response trajectory it records. The distinction between information exposure and questioning is important. Bail and colleagues experimentally exposed participants to messages from opposing political viewpoints and found that Republicans subsequently reported more conservative attitudes after following a liberal Twitter bot; the smaller change among Democrats exposed to conservative messages was not statistically significant [3]. That intervention supplied political content from ideological opponents. We ask whether selecting and sequencing *questions alone*, without providing persuasive arguments or recommended opinions, can produce measurable directional movement in an individual's survey-response profile.

This possibility motivates a different use of survey modeling. Rather than merely selecting questions for coverage, we would like to estimate how the response state would change under the possible answers to each candidate question, and then choose the next item by its expected directional effect. To do so, we infer conditional dependencies directly from survey data: for each item, we estimate its response distribution given the others. These are empirical associations, not an assumption that every answer influences all subsequent predictions. The fitted models provide a common distance between partial response states and a basis for ranking questions by their possible consequences. In principle, any method providing these conditional distributions could support the procedure; we use an LSM implementation.

Here we use one such implementation: a *Large Science Model* (LSM) composed of coupled conditional inference trees [4], [5]. Any variable can be a candidate predictor in other item-specific trees, allowing dependencies to be selected from the observed data rather than imposed. Given a subset of recorded answers, the remaining variables are left unobserved; native trees average over missing branches to calculate distributions for unasked items. These distributions and their uncertainty form the respondent's *predictive phenotype*. Thus the fitted model supplies a survey-derived digital twin for prediction without generating the participant's actual responses.

Related predictive digital-twin methods have been applied to the infant microbiome to predict neurodevelopmental deficits [6] and to influenza evolution for vaccine-strain forecasting and emergence-risk assessment through Emergenet [7]. These applications illustrate the use of data-derived dependencies for prediction. Here we use the learned conditional structure to *select* questions and test the direction of subsequent human response trajectories without delivering new political arguments.

The fitted conditionals then induce a geometry for comparing response states: states are close when the model predicts similar answer distributions for the remaining survey items. We average Jensen–Shannon-based discrepancies between these predicted distributions to define a distance measuring their *conditional consequences*, including for partially observed states. Whether an additional answer changes these predictions, and by how much, depends on the learned structure and the answer supplied. For the General Social Survey (GSS) 2024 model, we specify opposing Left and Right reference

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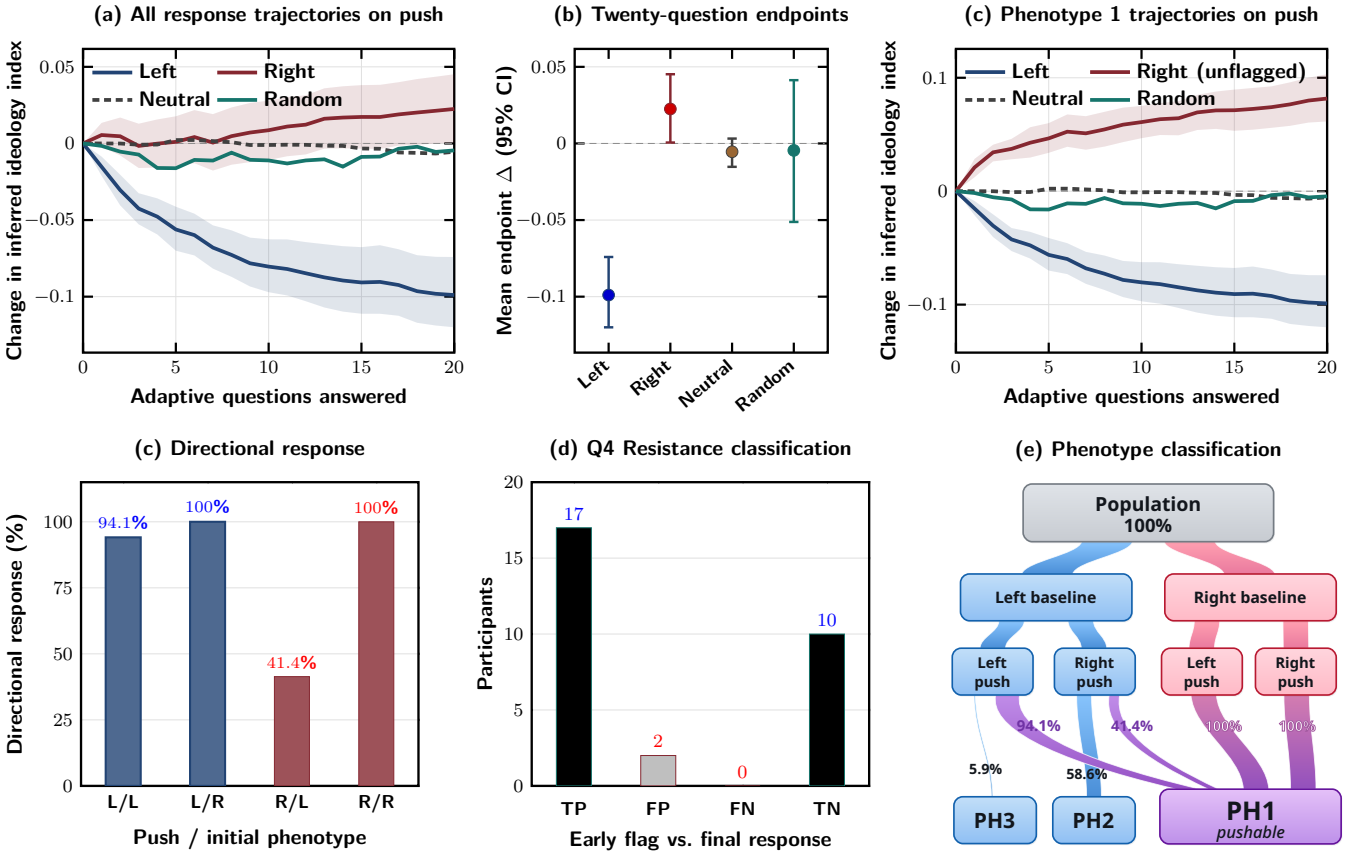


Fig. 1. **Human-study trajectories and push-response phenotypes (71 calibrated completers).** Top row, left to right: (a) mean baseline-relative trajectories under Left, Right, Neutral, and Random assignments, with 95% participant-bootstrap bands for Left and Right; (b) mean 20-question endpoint displacement with 95% participant-bootstrap intervals; and (c) an early-screened, PH1-enriched trajectory display. In (c), the Right-push curve retains the 19 participants not flagged at question four; Left, Neutral, and Random curves are unfiltered. This screening-defined subset is not the full endpoint-defined Right-push PH1 group ($n = 21$). Bottom row, left to right: directional success by push/baseline sign (L/L: 16/17, L/R: 4/4, R/L: 12/29, R/R: 9/9; the first symbol denotes the assigned push); four-question resistance-screening outcomes among initially Left-leaning Right-push participants (17 TP, 2 FP, 0 FN, 10 TN); and the operational phenotype classification. **PH1 (pushable phenotype)** denotes net movement in the assigned direction (41/59 directional-arm participants). **PH2 (Right-push resistant phenotype)** denotes no net rightward movement from a Left baseline (17/29, 58.6%). **PH3 (Left-push resistant phenotype)** denotes no net leftward movement from a Left baseline (1/17, 5.9%). All 13 participants with Right baselines responded in their assigned directions. PH1–PH3 classify participants' observed response trajectories using a fixed conditional-response distance derived from the survey-dependency model.

states on 27 survey items spanning policy, civil liberties, spending, and religion. The difference between a respondent's distances to these reference states gives a signed, pole-relative coordinate. As participants provide further answers, their observed response states are augmented and the coordinate is recalculated using the same fitted metric and reference poles. Any resulting displacement is measured from their actual answers, not from model-generated responses.

We test this reflexive possibility using *Digital Twin Anchored Generation* (DTAG), an adaptive question-selection procedure built on the inferred conditional dependencies. For each eligible unasked item, DTAG calculates the coordinate that would follow from each possible answer and weights these candidate displacements by the fitted conditional response probabilities. This gives a model-based expected signed displacement for selecting the next question; its realized effect depends on the answer received. A Left-directed policy selects the item with the most negative expectation; a Right-directed policy selects the most positive. Neutral and Random policies provide comparisons. The model determines *which question to ask*, not how the person will answer. Participants respond in their own words, which a language-model interface maps to native survey-variable assignments. The participant receives the question, not a curated political message, argument, or suggested answer. Each observed response augments the recorded state, and its distance-based coordinate is recalculated. Whether the response profile moves in the intended direction is measured rather than assumed. This setup tests directional inquiry without conflating it with the exposure interventions studied by Bail and others.

Among 71 baseline-calibrated human completers, 59 received 20 Left- or Right-directed adaptive questions. Leftward movement occurred in 20/21 Left-push participants (95.2%), compared with rightward movement in 21/38 Right-push participants (55.3%). Directional movement also opposed the initial orientation: all four initially Right-leaning participants assigned Left push moved left, while 12/29 initially Left-leaning participants assigned Right push moved right. The other 17 participants in that Right-push subgroup had no net rightward displacement; all 17 were flagged by their first four adaptive answers, along with two eventual responders. Baseline orientation is defined by the sign of the distance-based coordinate after six observed answers; success is endpoint movement in the assigned direction.

These trajectories define three operational response phenotypes: PH1, *pushable* in the assigned direction (41/59; 69.5%);

PH2, resistant to Right push from an initially Left-leaning baseline (17/29; 58.6%); and PH3, resistant to Left push from an initially Left-leaning baseline (1/17; 5.9%). All 13 initially Right-leaning participants moved in their assigned directions. The categories are based on observed endpoint displacement under the assigned policy; PH3 consists of one participant.

Work on attitude organization and partisan sorting motivates estimating cross-item associations beyond a single ideological dimension [8]–[11]; cognitive consistency may also influence successive reported attitudes [1], [2], [12]. Studies of political exposure and interpersonal contact document heterogeneous responses to new information or social encounters [3], [13], [14]. Unlike lower-dimensional opinion-dynamics models [15]–[17] or language-model-generated survey personas [18], [19], DTAG selects questions using a learned high-dimensional dependency structure, leaving every answer to the participant. We first characterize the human response trajectories and phenotypes, then specify the conditional geometry and selection procedure. Supporting computational tests comprise 960 model-sampled trajectories and 2,880 fixed-set replays of selected question sets.

2 Human Response Phenotypes

Of 71 calibrated completers, 59 received directional questioning and 12 received Neutral or Random policies. We quantify endpoint displacement, PH1–PH3, and early Right-push nonresponse screening (Fig. 1); Methods describes the protocol.

2.1 Outcome definitions and observed phenotype classes

Let s_0 denote the six-answer baseline and s_t the response state after adaptive answer t. Baseline-relative displacement is

$$\Delta I_\Phi(t) = I_\Phi(s_t) - I_\Phi(s_0), \qquad t = 0, \ldots, 20. \tag{1}$$

A directional success at the end of the adaptive sequence is $\Delta I_\Phi(20) < 0$ for Left and $\Delta I_\Phi(20) > 0$ for Right. For a directionally assigned participant, set $d = -1$ for Left push and $d = +1$ for Right push and define the signed endpoint response $R_d = d\,\Delta I_\Phi(20)$. We assign the following operational, policy-conditional response phenotypes:

- **PH1 (pushable phenotype):** $R_d > 0$, indicating net displacement in the assigned direction regardless of baseline sign.
- **PH2 (Right-push resistant phenotype):** $I_\Phi(s_0) < 0$, $d = +1$, and $R_d \leq 0$, indicating no net rightward displacement from an initially Left-leaning baseline.
- **PH3 (Left-push resistant phenotype):** $I_\Phi(s_0) < 0$, $d = -1$, and $R_d \leq 0$, indicating no net leftward displacement from an initially Left-leaning baseline.

Resistance denotes no net movement in the assigned direction after 20 questions, including zero displacement. PH1–PH3 exhaust this observed directional cohort because no initially Right-leaning participant resisted; they need not exhaust other samples. Neutral and Random participants are unclassified. The fixed LSM-defined distance measures changes in observed responses.

2.2 Completed sessions and directional response

The 71 calibrated completers comprise 25 from the original four-policy cohort, 15 fixed-Left recruits, and 31 fixed-Right recruits. Policies total Left $n = 21$, Right $n = 38$, Neutral $n = 7$, and Random $n = 5$. The separately recruited zero-calibration Right cohort is excluded for lack of a measured six-question baseline.

After 20 questions, 20/21 Left-push participants (95.2%) moved left and 21/38 Right-push participants (55.3%) moved right. Left-baseline response was 16/17 Left push (94.1%) and 12/29 Right push (41.4%); Right-baseline response was 4/4 Left and 9/9 Right (100% each). Of 59 directional participants, PH1 comprises $16 + 4 + 12 + 9 = 41$ (69.5%), PH2 comprises 17 (28.8% of the directional cohort, 58.6% of Left-baseline Right push), and PH3 comprises one (1.7% of the directional cohort, 5.9% of Left-baseline Left push). Resistance depends on both assigned policy and response, not baseline sign alone.

The mean endpoint displacement was -0.0989 for Left push (95% participant-bootstrap confidence interval $[-0.1200, -0.0741]$) and $+0.0225$ for Right push ($[+0.0006, +0.0452]$). Neutral and Random showed smaller mean displacements (-0.0054 and -0.0045, respectively).

2.3 Early identification of Right-push resistance

Among 29 initially Left-leaning Right-push participants, the question-four criterion flags no increase from baseline after four adaptive answers. Final PH2 status requires no net rightward movement after 20 answers. The flag therefore screens for PH2, but early-flagged participants can finish as PH1.

All 17 eventual nonresponders were flagged at question four, along with two eventual responders: sensitivity 17/17 (100%), precision 17/19 (89.5%), specificity 10/12 (83.3%), and accuracy 27/29 (93.1%).

2.4 Interpretation

PH1 occurred in 41/59 directional participants: all 13 initially Right-leaning participants and 28/46 initially Left-leaning participants (the latter pool combines different policies). Among Left-baseline participants, resistance was 17/29 (58.6%) under Right push (PH2) and 1/17 (5.9%) under Left push (PH3). Each participant received one policy; the same person's response to the opposite push remains unmeasured. All 17 PH2 outcomes, plus two eventual PH1 responses, met the four-question screen. Its accuracy in this cohort was 27/29; the rule can be applied after four answers, and independent validation is needed.

3 Methods

3.1 Human-study recruitment and response mapping

Prolific participants received six baseline questions, 20 adaptive questions, and five interleaved attention checks. Policies were Left, Right, Neutral, and Random: the initial cohort was randomized; later fixed-Left and fixed-Right cohorts were recruited separately. The LSM chose the question; participants supplied natural-language answers. A language-model interface presented questions and mapped each answer to one or more accepted native survey-variable assignments, updating the response state without generating the participant's answers.

3.2 Conditional-dependency model of survey responses

Let $X = (X_1, \ldots, X_n)$ denote categorical survey responses, with alphabets $\mathcal{A}_i$ and context c. We infer the response dependencies through the collection of conditional distributions

$$\Phi_c = \{\phi_{i,c}(a \mid x_{-i}) : i = 1, \ldots, n\}, \tag{2}$$

TABLE 1
Directional response after the 20 adaptive questions, stratified by the sign of the baseline inferred ideology index. Success denotes net displacement in the assigned direction.

Initial baseline index	Left push	Right push
Initially Left	16/17 (94.1%)	12/29 (41.4%)
Initially Right	4/4 (100%)	9/9 (100%)
Overall	20/21 (95.2%)	21/38 (55.3%)

TABLE 2
Human-study results by assigned policy. Endpoint displacement is measured from each participant's six-question baseline; intervals are 95% participant-bootstrap percentile intervals (5,000 replicates).

Policy	n	Baseline $I_\Phi < 0$	Baseline $I_\Phi > 0$	Mean $\Delta I_\Phi(20)$ (95% CI)	Directional success
Left	21	17	4	-0.0989 $[-0.1200, -0.0741]$	20/21 (95.2%)
Right	38	29	9	$+0.0225$ $[+0.0006, +0.0452]$	21/38 (55.3%)
Neutral	7	6	1	-0.0054 $[-0.0153, +0.0032]$	—
Random	5	2	3	-0.0045 $[-0.0512, +0.0413]$	—

TABLE 3
Early resistance classification among initially Left-leaning participants assigned Right push ($n = 29$). The classification uses the first four adaptive answers; the final outcome is measured relative to baseline at the end of the 20-question sequence.

	No net rightward move	Moved rightward
Flagged resistant	17	2
Not flagged	0	10

Each conditional estimates how one survey item's response distribution varies with the other recorded items. We fit these conditionals as a Large Science Model using conditional inference trees [4]–[6]. Whether and which predictors influence an item is learned from the survey data. Native tree averaging handles missing inputs; the fitted distributions define the predictive phenotype, measurement geometry, and question-selection objective.

The observed state $s = \{X_j = a_j : j \in S\}$ contains recorded responses. For an unasked item $v \notin S$, unobserved inputs remain missing in $\widetilde{x}_{-v}(s)$:

$$p_\Phi(a \mid s, v) = \phi_{v,c}(a \mid \widetilde{x}_{-v}(s)). \tag{3}$$

The vector

$$\mathcal{B}_\Phi(s) = \big(p_\Phi(\cdot \mid s, v)\big)_{v \notin S} \tag{4}$$

is the *predictive phenotype*: conditional distributions and uncertainty for unasked responses.

3.2.1 Predictive geometry and pole-relative index

Conditional predictions induce a distance between complete or partial states x, y:

$$\theta_\Phi(x, y) = \frac{1}{n} \sum_{i=1}^{n} \sqrt{D_{\mathrm{JS}}(\phi_{i,c}(\cdot \mid x_{-i}), \phi_{i,c}(\cdot \mid y_{-i}))}. \tag{5}$$

Here D_{JS} denotes Jensen–Shannon divergence. The resulting predictive pseudometric distinguishes states by their conditional consequences.

Prior LSM reconstruction reduced mean distance distortion by 54.9% (GSS) and 60.9% (Eurobarometer) with 80% of responses masked [5].

For GSS models with registered left and right reference states s_L, s_R, DTAG reports

$$I_\Phi(s) = \frac{\theta_\Phi(s_L, s) - \theta_\Phi(s_R, s)}{\theta_\Phi(s_L, s_R)}. \tag{6}$$

The reference poles specify opposing responses on 27 GSS 2024 items covering policy, civil liberties, spending, and religion (Table 4) [5]. Other coordinates are unspecified; I_Φ measures position in the predictive geometry induced by the LSM.

3.3 Adaptive selection from conditional response predictions

For each candidate question, we evaluate the coordinate changes under its possible answers and average them using the fitted conditional probabilities. This yields an expected geometric effect, not a fixed change caused by asking the question. In native simulation, a fixed persona anchors the state; the selected item v_t receives an LSM-sampled response:

$$A_t \sim p_\Phi(\cdot \mid s_t, v_t), \tag{7}$$

$$s_{t+1} = s_t \cup \{v_t = A_t\}, \tag{8}$$

$$\mathcal{B}_{t+1} = \mathcal{B}_\Phi(s_{t+1}). \tag{9}$$

In human sessions, observed answers replace model draws. A single free-text answer may map to multiple accepted native assignments; the equations describe one-item native updates. Question selection does not prescribe the answer.

For candidate item v and answer a, define

$$\Delta_v(a; s) = I_\Phi(s \cup \{v = a\}) - I_\Phi(s), \tag{10}$$

$$\mu_v(s) = \sum_{a \in \mathcal{A}_v} p_\Phi(a \mid s, v) \Delta_v(a; s), \tag{11}$$

$$\sigma_v^2(s) = \sum_{a} p_\Phi(a \mid s, v)(\Delta_v(a; s) - \mu_v(s))^2. \tag{12}$$

Right- and left-seeking policies maximize and minimize $\mu_v(s)$; neutral minimizes $|\mu_v(s)|$, and random samples an eligible item. By construction,

$$\mathbb{E}[I_\Phi(s_{t+1}) - I_\Phi(s_t) \mid s_t, v_t] = \mu_{v_t}(s_t).$$

Accumulated displacement follows successive selected items and their realized answers, which are LSM draws in simulation and participant responses in the human experiment.

TABLE 4
GSS 2024 reference poles. Twenty-seven paired responses define s_L, s_R, with $I_\Phi = -1, +1$; all other coordinates are unspecified.

Item	Survey topic	Left	Right
abany	Abortion on request	yes	no
abdefect	Abortion: fetal defect	yes	no
abnomore	Abortion: no more children	yes	no
abpoor	Abortion: low income	yes	no
abrape	Abortion after rape	yes	no
absingle	Abortion: unmarried	yes	no
bible	Bible interpretation	ancient book	inspired word
colcom	Communist as college teacher	not fired	yes, fired
confed	Trust in federal government	a great deal	hardly any
conlabor	Trust in organized labor	a great deal	hardly any
grass	Marijuana legalization	should be legal	should not be legal
gunlaw	Permit to purchase gun	favor	oppose
libcom	Communist books in library	not remove	remove
libmslm	Muslim cleric books in library	not remove	remove
natarms	Military spending	too much	about right
natenvir	Environmental spending	too little	about right
natfare	Welfare spending	too little	too much
natsoc	Social Security spending	too little	about right
owngun	Guns in household	no	yes
pillok	Contraception access	strongly agree	disagree
polabuse	Police strike insulting citizen	yes	no
pray	Frequency of prayer	never	several times a day
prayer	Bible prayer in public schools	approve	disapprove
religen	Personal religiosity	no religion	strong
rowngun	Respondent owns a gun	no	yes
shotgun	Owns a shotgun	no	yes
spkcom	Communist public speech	yes, allowed to speak	not allowed

The effect on an unasked item j after observing (v, a) is

$$R_j(s; v, a) = D_{\mathrm{JS}}\big(p_\Phi(\cdot \mid s, j), p_\Phi(\cdot \mid s \cup \{v = a\}, j)\big). \quad (13)$$

A weighted sum $\mathcal{P}_\Phi(s; v, a) = \sum_{j \notin S \cup \{v\}} w_j R_j(s; v, a)$ measures redistribution across the remaining items. Unlike text retrieval [20], DTAG maintains an explicit conditional response state. The public implementation includes a web application (https://github.com/zeroknowledgediscovery/DTAG/tree/main/webapp).

3.4 Phenotypic refinement and possible response effects

Two effects of questioning are conceptually distinct. First, conditioning the model on an observed answer can change predictions for unasked items, yielding *phenotypic refinement*. Second, posing a question may alter later human responses through salience, reflection, or response-order effects: *survey reflexivity*. The former is defined by the fitted conditional distributions; the latter concerns how human answers may evolve through inquiry. For held-out responses Y, the expected information from asking v is

$$\mathcal{G}(v \mid s) = H(Y \mid s) - \mathbb{E}_{A_v|s}[H(Y \mid s, v = A_v)]. \quad (14)$$

Under a coherent joint model this is conditional mutual information, distinct from directional displacement $\mu_v(s)$. Held-out response prediction evaluates refinement directly.

3.4.1 Possible response effects

DTAG elicits answers without supplying persuasive arguments or political messages. Nevertheless, the questions may influence what becomes salient and what participants subsequently report. A conceptual response–transition kernel separates the recorded answer from a possible response-state update:

$$K_q(a, z' \mid z) = O_q(a \mid z)\, T_q(z' \mid z, a). \quad (15)$$

Passive observation has $T_q(z' \mid z, a) = \delta_z(z')$; salience, reflection, or belief changes could instead yield nontrivial transitions. This kernel distinguishes possible reflexive response dynamics from conditional measurement updates; the human study measures the resulting response trajectories rather than estimating the transition kernel.

3.4.2 Order dependence

For fixed realized answers, the terminal index is order-independent; sequentially generated answers need not be. A coherent joint distribution satisfies

$$P(a \mid s, v)P(b \mid s, v = a, w) = P(b \mid s, w)P(a \mid s, w = b, v). \quad (16)$$

Separately fitted LSM conditionals may violate this identity [21]. Fixed-set replays evaluate its consequences for native response sequences.

3.5 Native-model computational validation

We test selection with answers sampled from the fitted conditional model.

3.5.1 Adaptive questioning from pinned persona states

We initialized six specified GSS 2024 personas (Table 5) and ran 40 independent 20-question trajectories per persona under Left, Right, Neutral, and Random policies. Answers were sampled from LSM conditionals, not generated by an LLM. All 27 pole-defining items were excluded (Table 4). Thus 960 trajectories produced 19,200 non-pole answers. Both directional policies moved the index as assigned across all six personas (Fig. 2); Neutral and Random supplied comparisons. Movement arose through non-pole items.

3.5.2 Question selection as observed paths

Question identities are summarized in Table 6. The 18 displayed heatmaps contain 161 distinct items across right, left, and neutral selection (Fig. 3). A shared categorical ordering approximately minimizes

$$\mathcal{C}(\pi) = \sum_{p,u,t} \sum_{r<r'} \big|\pi(v_{purt}) - \pi(v_{pur't})\big|, \quad (17)$$

Here u indexes policies. Heatmap colors identify items, not ideological position; 67 further items occurred only in random selection.

3.5.3 Fixed-set replay: robustness to question ordering

For each persona and Left/Right/Neutral policy, we froze the 20-item sequence from adaptive replicate zero (18 sets). Each set was replayed in original, reverse, and two shuffled orders, with 40 new LSM response sequences per order (2,880 runs). Matched question-specific random variates reduced contrast variance. Right-derived means were +0.0381 (original) and +0.0377 (reverse); Left-derived means were −0.0399 and

TABLE 5
Pinned GSS 2024 synthetic personas and baseline pole-relative index.

Persona	Pinned description	Initial $I_\Phi(s_0)$
Alabama	45-year-old rural Alabama white man; married, two children; full-time farmer and veteran; high school; conservative, strong Republican; Protestant, weekly attendance; daily newspaper.	0.075706
New York	22-year-old New York City white woman; unmarried, no children; full-time retail worker, nonveteran; bachelor's degree; extremely liberal, strong Democrat; nonreligious; daily newspaper.	−0.163807
Texas	61-year-old small-town Texas white woman; married, three children; homemaker, high school; extremely conservative, strong Republican; Protestant, attends services several times weekly; daily newspaper.	0.045379
Seattle	34-year-old Seattle white man; unmarried, no children; employed full time, graduate degree; liberal, weak Democrat; nonreligious; newspaper several times weekly.	−0.119639
Arizona	39-year-old Mexican American man in an Arizona city; married, three children; employed full time, high school; conservative, weak Republican; Catholic, nearly weekly attendance.	−0.020437
Atlanta	47-year-old Atlanta Black woman; divorced, two children; employed full time, bachelor's degree; liberal, strong Democrat; Protestant, weekly attendance; daily newspaper.	−0.030807

Positive values lie nearer the right reference pole; negative values nearer the left.

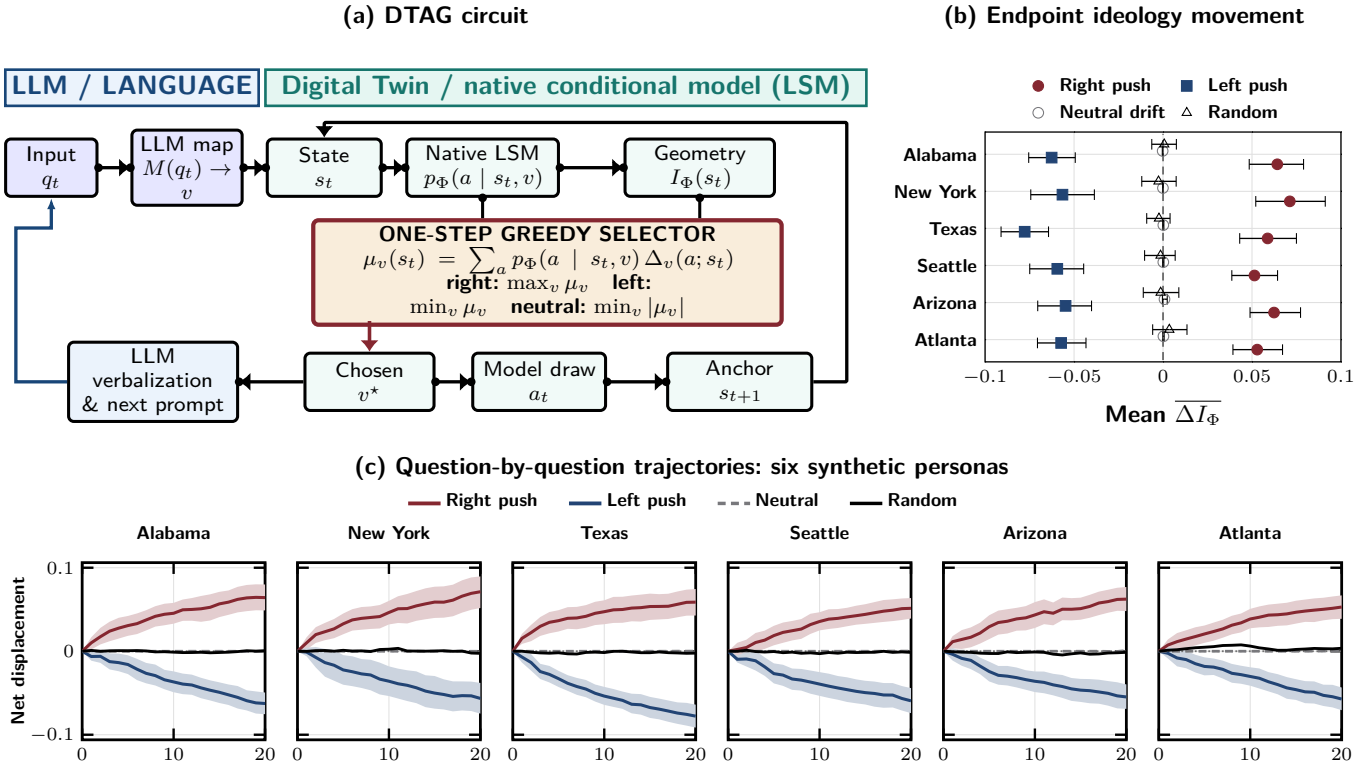


Fig. 2. **DTAG architecture and native adaptive results.** (a) LSM-anchored responses and directional question selection; language mapping is a separate interface. (b) Mean endpoint displacement across six personas with 95% bootstrap intervals. (c) Mean 20-question trajectories over 40 repetitions per persona and policy, with pointwise 95% intervals (right: red; left: blue; neutral: dashed gray; random: black).

TABLE 6
Question selection in the native GSS 2024 experiment.

Selection policy	Distinct	Selections	Pole overlap
Right-seeking	100	4,800	0
Left-seeking	75	4,800	0
Neutral	43	4,800	0
Three-policy union	161	14,400	0
All four policies	228	19,200	0

The random policy contributed 4,800 selections; 67 of the 228 distinct items occurred only under random selection.

−0.0410. All six persona-specific 95% bootstrap intervals excluded zero in the assigned direction for both original and reverse orders. Shuffles preserved the direction; Neutral sets remained near zero (Fig. 3). All 54 pointwise bootstrap intervals for paired mean between-order contrasts included zero: no systematic mean order effect was detected, and the question sets retained direction without continuous reoptimization.

4 Discussion and Conclusion

Adaptive question selection yielded systematic directional movement in survey-response states built entirely from human-supplied answers. Unlike experiments that expose participants to opposing political information, DTAG introduces no persuasive argument or recommended opinion: participants receive questions and choose their own responses. The observation is therefore a form of *directional inquiry* that raises a reflexive

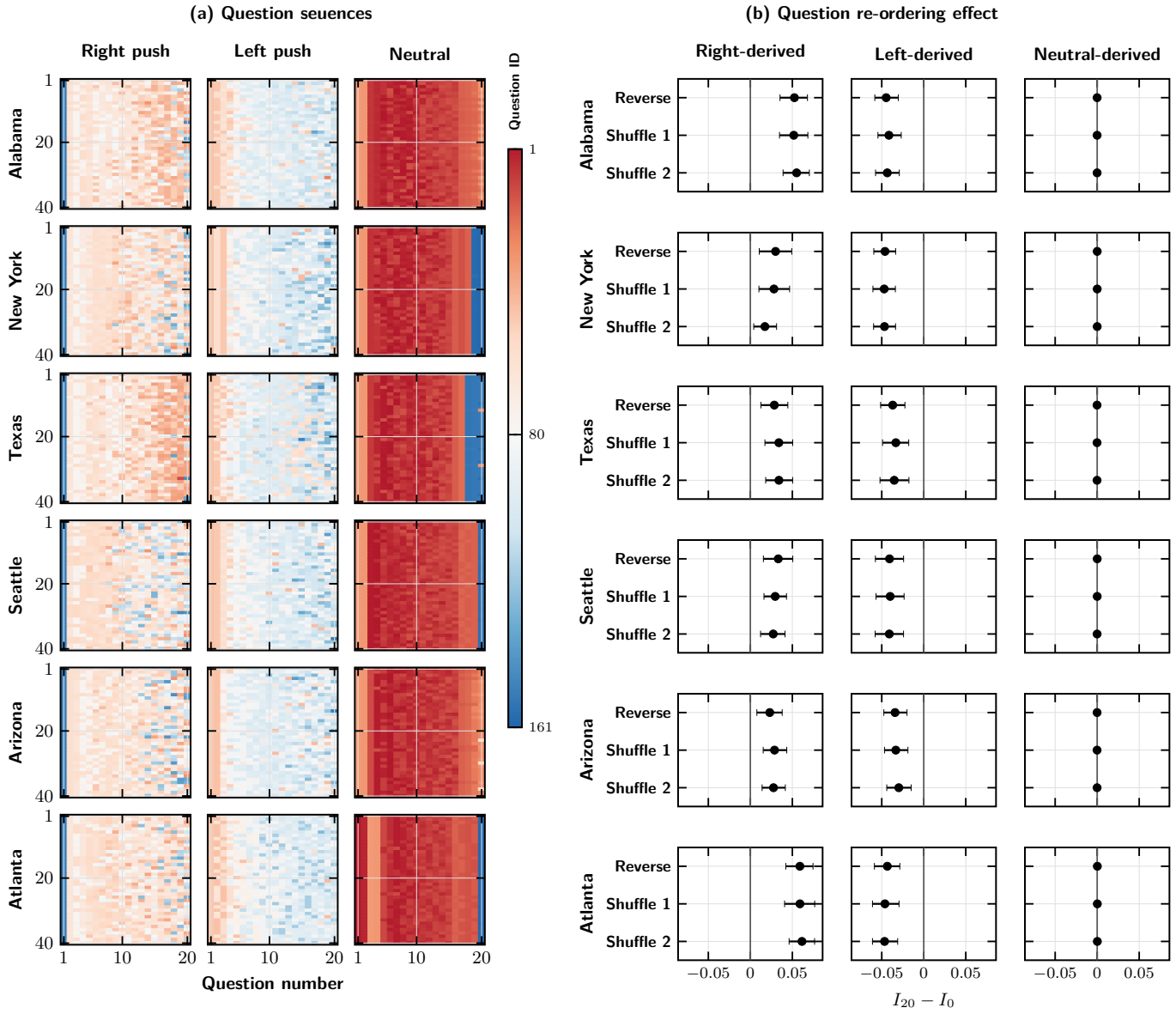


Fig. 3. **Selected questions and fixed-set replays.** Left: question-ID heatmaps for six personas and three selection policies (40 runs each); color denotes item identity. Right: mean index displacement and 95% bootstrap intervals for 20-item sets replayed in original, reversed, and two shuffled orders. Right- and left-derived sets retain their direction; neutral sets remain near zero.

question: how much can the selection of observations shape the trajectory that a survey records?

This distinction is illustrated by Bail et al., who found that Republicans exposed to liberal political messages subsequently reported more conservative attitudes, whereas the corresponding Democratic effect was not statistically significant [3]. Their manipulation involved new political content. DTAG manipulates question selection, not exposure to curated political arguments. The observed trajectories could reflect the salience and reflection elicited by questioning, as well as the information contained in the answers. The experiment measures the net displacement in a fixed conditional-response geometry without requiring the mechanism to be assigned in advance.

The directional asymmetry was pronounced: 20/21 Left-push participants moved left, compared with 21/38 Right-push participants moving right. Movement could oppose baseline orientation (4/4 initially Right-leaning under Left push and 12/29 initially Left-leaning under Right push), while Neutral and Random mean displacements were smaller (-0.0054 and -0.0045). PH1 comprises 41/59 directionally responsive participants. Among Left-baseline participants, 17/29 resisted Right push (PH2), but only 1/17 resisted Left push (PH3). All nine Right-baseline participants assigned Right push moved right. The aggregate Right-push result therefore conceals distinct responsive and resistant subgroups.

The first four answers already distinguished PH2: all 17 eventual PH2 outcomes were flagged, with two false positives. Within-cohort sensitivity was 17/17 and accuracy 27/29. The flag can be computed at question four; independent validation remains necessary. PH3 was observed in one participant.

The model-based experiments test the question-selection mechanism separately. In 960 LSM-sampled trajectories, the selection rule produced directional displacement without choosing any of the 27 pole-defining items. In 2,880 fixed-set replays, the direction persisted across original, reversed, and shuffled orders, with no systematic mean order effect detected in the tested comparisons. Unlike scripted persuasion or language-model-generated survey personas [18], [19], the

human trajectories use answers supplied by participants.

These results position conditional-dependency- guided questioning as a tool for investigating the reflexivity of survey measurement and direction-specific heterogeneity. The fixed metric quantifies changes in observed response states; subsequent independent or delayed measurement would distinguish persistent changes in reported attitudes from changes revealed by accumulating answers. The original four-policy cohort was randomized, while later fixed-policy cohorts were recruited separately; each participant received one assigned policy. The four-question rule can be applied before the final outcome is known and merits validation in an independent cohort. The general methodological principle extends to multivariate systems where learned conditional structure permits meaningful state comparisons and targeted inquiry.

5 Scope and Limitations

The LSM defines the fixed distance metric and selection expectations; participants supply observed answers. The protocol lacked delayed or independent response retesting to distinguish persistent changes in expressed attitudes from sequentially revealed responses. Separately estimated conditionals need not define a coherent joint distribution [21]. Pooled results combine randomized and separately recruited policy cohorts. Participants experienced one directional policy each, and PH3 has one participant. The four-question PH2 screen uses only answers available at question four; its reported performance was evaluated against final outcomes in the same 29 Left-baseline Right-push participants and needs independent validation. Fixed-set replays tested specific orderings, not every possible sequence.

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